



An Indirect Impedance Eduction Process for Liners with Arbitrarily Complex Geometry

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Complex aero-engine liner geometries that extend the capabilities of standard 1- or 2-degree-of-freedom liners became very popular in recent years. The challenge when trying to characterize these devices in terms of an equivalent impedance is that the classical *in-situ* method to educe impedance from measured pressure fluctuations cannot be employed straightforwardly. The current study proposes an impedance eduction method based on a combined usage of a 3D scale-resolving flow simulation of a limited portion of the liner with a 2D frequency-domain FEM acoustic calculation in which the liner is replaced by an equivalent impedance boundary condition. Both simulations setups are generated by a fully automatic workflow based on specific user settings. The FEM calculations are driven by an optimization algorithm that changes the value of wall impedance until acoustic properties of the reference 3D solution are matched.

I. Nomenclature

P	=	complex pressure
F	=	complex Transfer Function
N	=	number of probes
i	=	probe index
Err	=	error
L	=	liner length
c	=	sound speed
M	=	Mach number
$\bar{f}$	=	wavepacket frequency
Z	=	complex impedance
R	=	resistance
X	=	reactance
r	=	non-linear resistance coefficient

II. Introduction

Recent developments of higher bypass-ratio and geared turbofans with lower fuel consumption resulted in a shift of fan noise towards lower frequencies. In order to preserve the noise reduction benefits associated with these modern engine architectures and secure the aircraft noise certification, the frequency of maximum absorption of the acoustic treatments lining the engine nacelle must be reduced accordingly. To achieve this frequency shift without increasing the liner thickness and thus further increasing the nacelle diameter and weight, new liner concepts are needed, which overcome the state-of-the-art technology of multi-layer honeycomb liners. This explains the renewed interest of the aeroacoustic research community in the development of acoustic liners for aircraft engines.

Thanks to the progress in manufacturing process via 3D printing techniques, some manufacturing barriers became less insurmountable, thus opening the way to new conceptual designs based on complex geometries, such as multiple resonators in a labyrinth layout, layers with a random cavity depth distribution, face sheets with slanted orifices, etc. In

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this context, scale-resolving CFD simulations are used to better understand the underlying physics of sound attenuation and drag generation, the latter being another important driver of research in new liner technologies. With the increase of computational power and software automation, CFD is also becoming a powerful design tool used to analyze the impact of design changes on the acoustic response. Among other methods, the Lattice-Boltzmann Very-Large-Eddy Simulation Method (LBM-VLES) implemented in the SIMULIA PowerFLOW® software has proven to be able to predict the expected sound absorption of micro-perforated resonators. The literature in this field covers different fundamental aspects, such as the effects due to orifice and cavity geometry [1], to grazing flow [2], to heating [3]. Attempts to compute fan noise generation and radiation from a lined nacelle have been also accomplished [4].

The acoustic response of liners is typically expressed in terms of the complex impedance $Z(f) = R + Xj$, a quantity that can be used to define an equivalent boundary condition in FEM models of sound propagation [5]. Impedance is not measured directly, but it is “educated” from measurements of pressure fluctuations, which are related to the liner impedance through a defined acoustic model. The simplest method proposed by Dean [6] consists in using pressure fluctuations on the facesheet and backplate surfaces, and relate them through a one-dimensional acoustic representation of the cavity. Such surfaces cannot be easily identified in new complex liner geometries, posing a challenge in terms of liner impedance characterization, and thus leading to the development of new impedance education techniques. These are usually hybrid in nature, using experiments conducted on a limited section of the liner to generate a dataset that works as reference while tuning an equivalent mathematical model in which the liner is modeled directly via an impedance boundary condition. Several existing acoustic propagation are generally used to solve this reduced model’s acoustics [7].

In the present study, we propose a fully simulation-driven approach to evaluate the equivalent impedance of a liner with arbitrarily complex geometry. A demonstration of the approach for a 1-DoF constant depth liner from literature is carried out. The numerical impedance education approach consists in two stages. First, a 3D transient flow simulation of an acoustic wave in a grazing flow impedance tube is carried out using the LBM solver PowerFLOW to obtain a reference dataset. This replaces the experimental test used in hybrid methods. Best practices developed in the past years are employed to properly resolve wall turbulence and flow features in the orifices. Secondly, a reduced 2D model of the same setup that uses an impedance boundary condition to model the liner is automatically generated and linked to a SIMPLEX optimization algorithm. This iterates a series of quick FEM acoustics simulations in the frequency domain performed with *OptydB-GFD* [5, 8], by adjusting the impedance boundary condition until the FEM solution matches the acoustic properties of the reference LBM solution.

III. Liner geometry and channel Layout

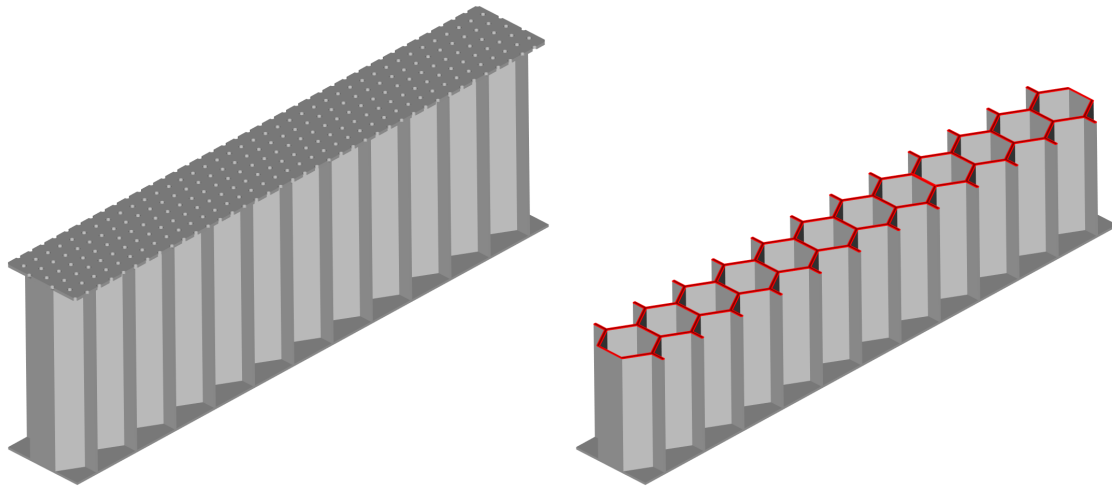


Fig. 1 Liner Geometry: full periodic geometry (left) and horizontally clipped geometry to show core honeycomb layout (right)

The liner lies on the floor of a flat channel. The conventional liner GE01 from [7] is used for this study (Fig. 1). This is a 1-dof perforate-over-honeycomb layout, using a 0.635 mm thickness plate with 0.991 mm diameter holes for 8.7% porosity. Core cells dept is 38.18 mm. Channel width is equal to the minimum periodic width unit of the liner

geometry (the process requires a minimum of one core cell span-wise). Periodic boundary conditions are applied along the channel span-wise direction. Liner length covers 12 core cells stream-wise. Channel height is set to 63 mm and channel length has been extended upstream and downstream of the liner to include a forcing wave-packet and sponge regions. The process can handle cases with or without grazing flow. For grazing flow cases, a solid trip on the channel

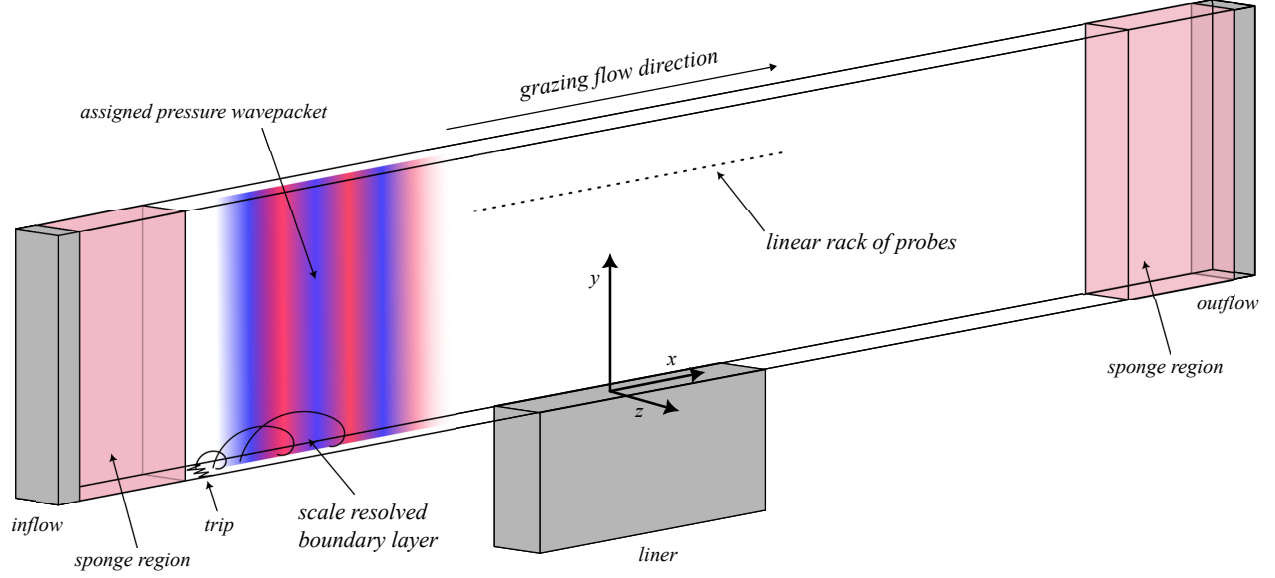


Fig. 2 Schematics of the 3D model (actual proportions not matched)

flow is used to trigger the direct solution of near wall turbulence. The trip distance upstream of the liner is automatically set to match a given boundary layer velocity profile or thickness at the liner which is specified by the user. Both 3D and 2D simulation models use the same layout. 2D models represent the liner only by means of a linear patch with an assigned impedance boundary condition.

IV. 3D solution

The first step in the impedance eduction process consists in generating the reference pressure dataset. In presence of grazing flow, an additional simulation is run at first to obtain a fully developed boundary layer in the channel. Once a regime flow solution is reached, simulation is halted. If grazing flow is absent, this simulation run for a single timestep in order to obtain the initial 3D fluid domain. In this initial field, a wave-packet with fixed frequency $\bar{f}$ is added via *OptydB-FIELDMOD* upstream of the liner and constitutes the initial condition for the following acoustic 3D simulation. The 3D acoustic run is conducted in the time domain with PowerFLOW and collects pressure data on a linear rack of N probes placed along the channel XY plane (channel mid-plane with $Z=0$) which covers the liner full length stream-wise. The number of probes and the position and extension of the liner rack can be modified via the process input parameters. Multiple racks at different heights are also supported. For the present study, a single linear rack of 100 probes placed at 55 mm height with upstream/downstream extensions of 10% of the liner length is used. The simulation run for a total duration that covers the passage of the entire wave-packet over the liner. Currently, the process uses a wave-packet with a single frequency and a new 3D acoustic run is needed for each frequency value under analysis. Top and bottom channel walls are treated as no-slip walls when a grazing flow is absent. When present, only the bottom wall starting from the trip location is modeled as a no-slip wall.

V. 2D solution

Once the 3D reference dataset is produced, the liner impedance is obtained by running several acoustics simulations in the frequency domain using a lightweight 2D model with similar channel layout. The liner is modeled via a simple boundary patch with location and extension that match the 3D model liner. An Ingard-Myers impedance boundary condition is used. The impedance of this patch is modified iteratively via an optimization scheme until a match between the 3D and 2D simulation is found.

For the solution of this reduced model, *OptydB-GFD* acoustics solver is used. This is a frequency-domain FEM/GFD solver. It supports two propagation models, Pierce's 2nd order (used in this study) and Lilley's 3rd order wave models [9]. It can be used for duct acoustics and fan noise radiation in the presence of lined walls [10–12], and to radiate the noise computed by rotors in the time domain, thanks to a specific wave-splitting technique [13].

For cases with grazing flow, the time-averaged flow field data obtained from the 3D simulation is interpolated onto the 2D mesh and accounted for in the solution. This permits to include spatially variable fluid material properties in the acoustic solution without the need of constant flow assumptions or simplified shear flow models.

In order to include, to some extent, non-linearity effects in the impedance boundary condition in 2D calculations, a new input parameter was added to *OptydB-GFD* solver. This defines an additional liner resistance coefficient proportional to the first derivative of velocity in the liner patch boundary condition.

VI. Mesh Resolution

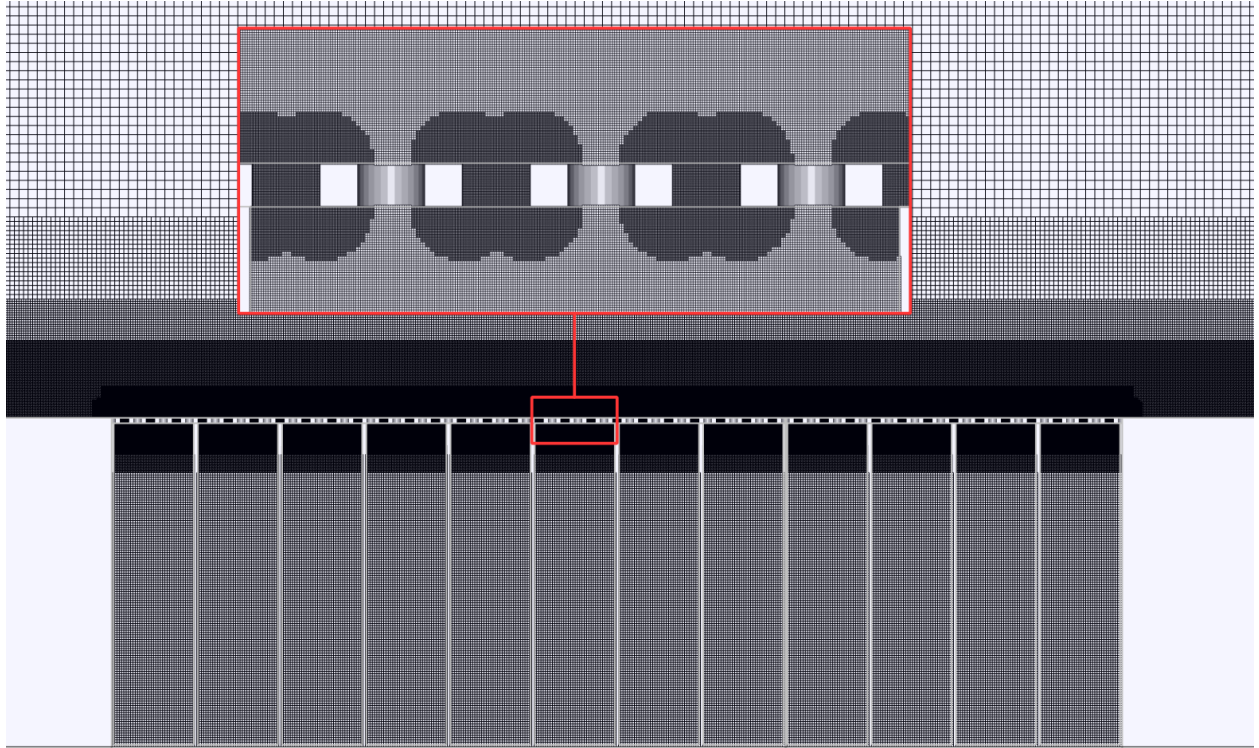


Fig. 3 PowerFLOW mesh (grazing flow case)

Several criteria are used in the 3D simulation to define local mesh resolutions. The first criteria sets the minimum voxel size threshold in order to have at least 30 voxels along the diameter of a single perforated plate hole or along the plate thickness, whichever is smaller (this criteria is based on previous works on liners with perforated plates, see [1] and [2]). The second criteria defines the mesh refinement around the solid trip, if present. This is equal to one third of the trip height, which is set targeting a Reynolds number based on the flow velocity and trip height of 2000. The third criteria is used to define the refinement near the bottom wall in presence of grazing flow. This is set to achieve $y^+ = 50$ at the wall. The forth and last criteria limits the size of the coarser mesh in the whole channel, ensuring that at least 15 mesh points are used along the maximum frequency wavelength (considering the flow convective velocity). Once all criteria thresholds have been established, the process defines first the coarser domain seed in order to have an even number of voxels along the liner width (a minimum of 2 voxels must be used). This is mandatory to ensure a correct domain periodicity span-wise in PowerFLOW. The actual mesh size follows by locally dividing the coarse seed mesh multiple times until each region mesh size threshold is reached. This leads to use roughly 40 voxels along the liner plate for this study, which corresponds to a linear resolution of 60 voxels/mm.

For the 2D simulation instead, a uniform mesh is used with size thresholds based on criteria that use the minimum wave-packet frequency (at least 20 mesh point per acoustic wavelength), the channel height (at least 64 mesh point) and the liner length (at least 100 mesh points). The smaller mesh size returned by these criteria is used to generate the 2D mesh (a mesh size of 1.2 mm is used herein).

VII. Process Automation

Parameter Name	Description
geometry scale	units used for the liner model (mesh)
Mach number	channel flow Mach Number. Setting this to 0 removes grazing flow settings
number of waves	number of waves in the wave-packet
frequency	frequency of the wave-packet
wave intensity	intensity [dB] of the wave-packet
min frequency	minimum wave-packet frequency in the study
max frequency	maximum wave-packet frequency in the study
temperature	flow temperature
pressure	flow pressure
viscosity	flow viscosity
channel height	height of the channel
orifices diameter	liner orifices diameter
liner geometry file	path to the liner model file
trip geometry file	path to the trip model file
boundary layer thickness	boundary layer thickness at the liner. Setting a negative value forces the process to search for a velocity profile to match within the project folder
flow sim auto-stop	flag to activate the auto-stop feature in PowerFLOW to automatically end the preliminary flow simulation in presence of grazing flow

Table 1 Process input parameters list

Parameter Name	Description
linear racks heights	height of the linear probe rack. Multiple values can be used to specify multiple probe racks
number of probes per rack	number of probes per single rack (equally spaced streamwise)
linear racks overlap extension	extension of the liner probe racks upstream and downstream of the liner as a percentage of the liner length
impedance initial guess	initial liner impedance guess value (complex)
non linear resistance initial guess	2D reduced model liner boundary condition non linear term initial guess value

Table 2 Post-processing input parameters list

The 3D/2D simulations setup and post-processing is fully automated. In order to run the process, the input parameters listed in Table 1 are set in a text input file. Processing of the results is also controller by setting the parameters listed in Table 2 in a second text file.

A python script launches the software SIMULIA PowerCASE® to generate the input file for PowerFLOW. The liner is aligned and positioned inside the channel, which is generated from scratch. Other entities required for the simulation are generated based on the input flow conditions. Minimum and maximum frequencies of interest are used to define the

length of the channel and the thickness of the sponge regions to avoid acoustics reflections. The wave-packet frequency selected controls instead the measurements sampling period and the total simulation length.

In presence of grazing flow, the distance of the trip upstream of the liner is automatically determined based on a given time-average near-wall velocity profile or alternatively on the desired boundary layer thickness at the liner. In this calculation, empirical turbulent boundary layer velocity profile expressions available in literature are used.

Once the 3D simulation is complete, the process can be run in post-processing mode to generate the reference dataset and build the equivalent 2D reduced model for *OptydB-GFD*. Pressure data probe signals from the 3D simulation are processed via SIMULIA PowerACOUSTICS®. Once the reduced model is built, the impedance eduction loop launches.

VIII. Optimization process

The optimization loop uses a SIMPLEX algorithm wrote *ad-hoc* for this study. The algorithm searches for the best match between 2D and 3D models results within the 3-parameters solution space of the liner patch resistance (R), reactance (X) and non-linear resistance coefficient (r).

The liner impedance is obtained by minimizing the following error metric between the transfer functions F of the complex pressure at probes locations obtained from the 3D (F_{BLM}) and 2D (F_{FEM}) simulations:

$$Err = \frac{1}{N} \sum_{i=1}^N \sqrt{(F_{iBLM} - F_{iFEM})^2}, \quad (1)$$

where the complex transfer function F is defined as:

$$F_i = \frac{P_i(\bar{f})}{P_1(\bar{f})} \quad (2)$$

The process is deemed converged when the error drops below a given threshold. The full algorithm used in this study is presented in the Appendix.

IX. Results

The impedance eduction is run for two flow conditions, without and with grazing flow, and for three values of the wave-packet frequency $\bar{f}$ of 1000 Hz, 1400 Hz and 1800 Hz. In all considered cases, the traveling wave-packet is set to 130 dB and includes 15 waves.

For grazing flow cases, Mach Number is set to 0.31, similarly to the reference experiment [7]. Fig. 4 shows the stream-wise velocity profile comparison with the experimental one [7]. In PowerFLOW, only the channel floor is modeled as no-slip wall to reduce the computational cost of the flow simulation. Despite the lower stream-wise velocity close to the wall, the boundary layer thickness is in good agreement with the reference data (roughly 25 mm).

Fig. 5 summarizes the impedance results. A comparison between the current study results and [7] is shown. In [7], two acoustic propagation models are used. The first assumes a finite length liner on one side of the channel and solves the convected Helmholtz equation via a finite element method approach (CHE-FE). It can include shear-layer effects modifying the original Helmholtz equation, but only results using a uniform velocity are reported for liner GE01. The second propagation model assumes a liner with infinite length and solves the Pridmore-Brown equation for the 1D pressure $P(y)$. Results considering both uniform velocity (PBU-SM) and 1D shear flow (PBS-SM) are reported.

Cases without grazing flow (black) are plotted together with grazing flow cases (red) when comparing with CHE-FE method results, while only grazing flow cases are compared with PBU-SM and PBS-SM approaches. In general, cases without grazing flow shows a very good agreement both in terms of resistance and reactance. Cases with grazing flow shows an overestimation of resistance when comparing with the CHE-FE approach, but are closer to the values predicted by the PBS propagation method, especially to PBS-sM for frequency $\bar{f} > 1000$ Hz. This suggests that the method is able to correctly capture the effect of the non-uniform flow field within the channel, which is not negligible. The case with grazing flow at $\bar{f} = 1000$ Hz shows a resistance value in between the two PBS approaches, and a reactance value that is smaller than all other cases, but respects the same trend when switching from no-flow to grazing flow case. It should be pointed out that, in the present study, a 2nd order wave model for the acoustic velocity potential has been adopted in the FEM calculation.

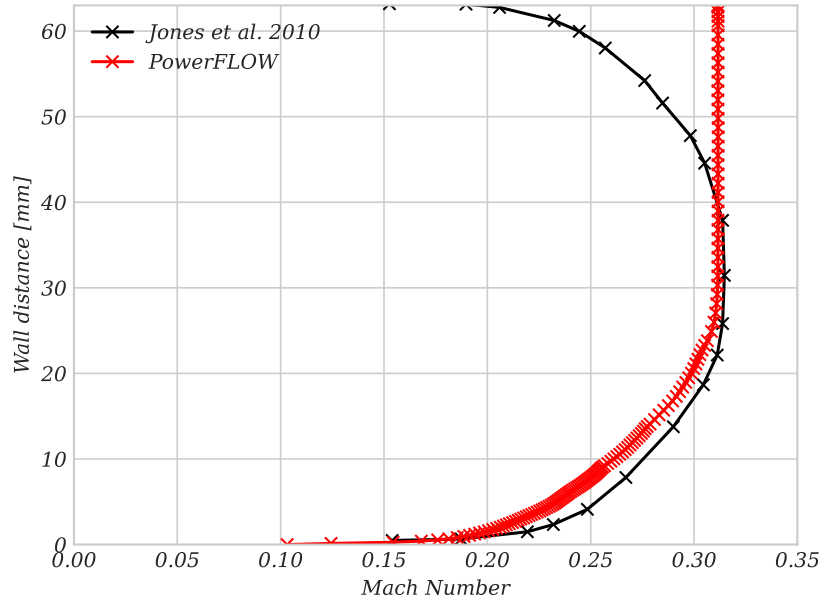


Fig. 4 Average velocity profiles comparison between [7] (black) and the present study (red) for cases with grazing flow. Reference Mach Number is 0.31

Conclusions

We presented a fully automatic process that uses 3D PowerFLOW simulations as reference and adopt an optimization loop using quick 2D FEM/GFD solutions of a reduced model to educt the equivalent impedance of a conventional liner. The process is completely simulation driven and applicable to arbitrarily complex liner geometries. Users are able to obtain the liner impedance by providing as inputs a periodic section of the 3D liner geometry and a series of test flow conditions.

In this study, a wave-packet at 130 dB with frequency ranging from 1000 Hz to 1800 Hz was used. Results for cases with and without grazing flow were compared with available literature data obtained with hybrid impedance eduction methods used at NASA [7]. A very good agreement was observed for cases without grazing flow. When a grazing flow is present, the predicted resistance is close to the literature reference when a propagation model that assumes a liner with infinite length and non-uniform flow (1-D sheared flow) is used. This suggests how the use of the full time-averaged flow field from the 3D simulation in our method is beneficial to the capability of capturing accurately non-uniformity effects of the material in the acoustic propagation calculation. From this point of view, our method ensures a more rigorous consistency between the reference 3D dataset and the reduced 2D model used for the eduction.

It is more difficult instead to comment about the discrepancy that occurs for $\bar{f} = 1000$ Hz. At low frequency, the limited length of the liner together with the larger wavelength of the wave-packet could play a role in making the liner boundary effects dominant. This could potentially reduce the capacity for our methodology to educt an impedance representative of a liner with the same periodic core but infinite length, but could be easily addressed by using a longer liner 3D section. Further analyses in which the length of the liner has been extended up to 24 cells are currently ongoing.

Tests with different liners models and wave-packet frequency ranges and intensities are planned. In future studies, the effect of using a 3rd order Pridmore-Brown wave model in the FEM calculation will be also analyzed.

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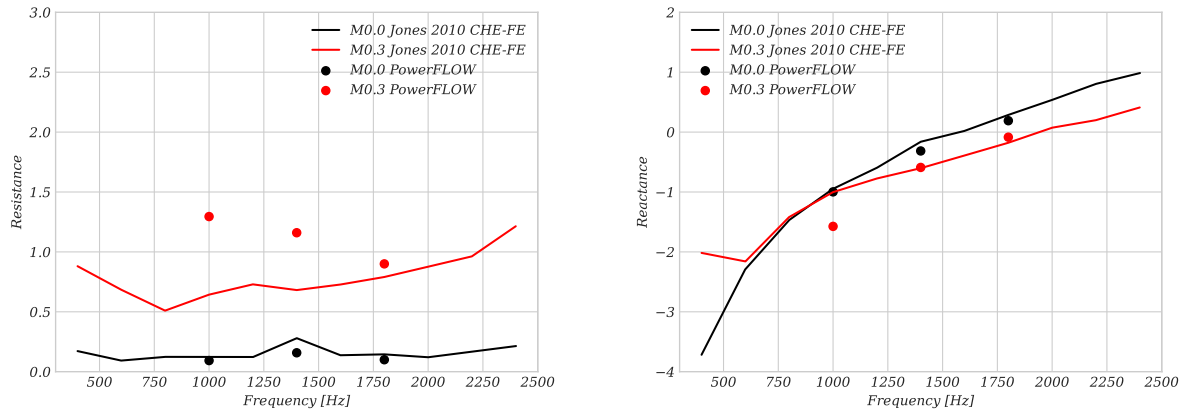


Fig. 5 Impedance comparison in terms of resistance (left) and reactance (right). No flow (black) and grazing flow cases (red). Results from [7] using CHE-FE approach (lines) and the present study (points)

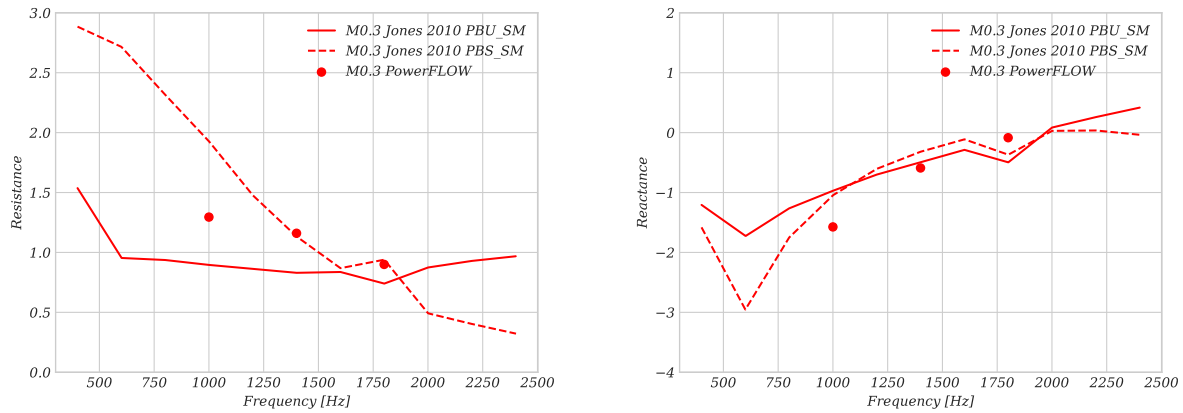


Fig. 6 Impedance comparison in terms of resistance (left) and reactance (right). Grazing flow cases only. Results from [7] using PBU-SM (continuous lines) and PBS-SM approach (dashed lines) and the present study (points).

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A. Appendix

The SIMPLEX optimization algorithm used in this study is reported herein (Algorithm 1). This minimizes the error between the pressure transfer functions of 3D and 2D simulations by modifying 3 parameters: the liner resistance (R), the liner reactance (X) and the non-linear resistance coefficient (r).

Convergence plots produced by the process are also shown. Fig. 7,8 show convergence plots for all cases. The convergence history charts on the left show how the 3 optimization parameters (impedance is here reported in term of admittance) are changed by the SIMPLEX algorithm during the eduction process. On the right instead a visual comparison of the reference (3D PowerFLOW) and reduced model (2D FEM) final transfer function is presented. Generally, a good match was obtained for all considered cases. This confirms the consistency existing between the 3D and 2D model responses.

Algorithm 1 3-parameters SIMPLEX algorithm

```

 $(R_1, X_1, r_1) \leftarrow (R_0, X_0, r_0)$  ▷ initial SIMPLEX stencil
 $(R_2, X_2, r_2) \leftarrow (R_0/2, X_0, r_0)$ 
 $(R_3, X_3, r_3) \leftarrow (R_0, X_0/2, r_0)$ 
 $(R_4, X_4, r_4) \leftarrow (R_0, X_0, \max(r_0 * 1.5, 0.001))$ 
 $err_1 \leftarrow \text{Opty}\partial B\text{-GFD}(R, X, r)$ 
 $err_2 \leftarrow \text{Opty}\partial B\text{-GFD}(R/2, X, r)$ 
 $err_3 \leftarrow \text{Opty}\partial B\text{-GFD}(R, X/2, r)$ 
 $err_4 \leftarrow \text{Opty}\partial B\text{-GFD}(R, X, \max(r * 1.5, 0.001))$ 
 $ERR \leftarrow \text{sort}([err_1, err_2, err_3, err_4])$ 
 $SIMPLEX \leftarrow \text{sort}([(R_1, X_1, r_1), (R_2, X_2, r_2), (R_3, X_3, r_3), (R_4, X_4, r_4)]) \text{ with } ERR$ 
while  $ERR[0] > \text{threshold}$  do ▷ optimization loop starts
     $B \leftarrow SIMPLEX[0]$ 
     $B.err \leftarrow SIMPLEX[0].err$ 
     $NB \leftarrow SIMPLEX[2]$ 
     $NB.err \leftarrow SIMPLEX[2].err$ 
     $W \leftarrow SIMPLEX[3]$ 
     $W.err \leftarrow SIMPLEX[3].err$ 
     $C \leftarrow \sum_{i=1}^3 SIMPLEX[i]/3$ 
     $R \leftarrow C + C - W$ 
     $R.err \leftarrow \text{Opty}\partial B\text{-GFD}(R)$ 
    if  $R.err \leq W.err$  &  $R.err \geq B.err$  then ▷ expansion
         $SIMPLEX[3] \leftarrow R$ 
         $ERR[3] \leftarrow R.err$ 
    else if  $R.err < B.err$  then ▷ extension
         $E \leftarrow CEN + 2 * (CEN - W)$ 
         $E.err \leftarrow \text{Opty}\partial B\text{-GFD}(E)$ 
        if  $E.err < R.err$  then
             $SIMPLEX[3] \leftarrow E$ 
             $ERR[3] \leftarrow E.err$ 
        else
             $SIMPLEX[3] \leftarrow R$ 
             $ERR[3] \leftarrow R.err$ 
        end if
    else if  $R.err > W.err$  then ▷ contraction
         $C \leftarrow (W + CEN)/2$ 
         $C.err \leftarrow \text{Opty}\partial B\text{-GFD}(C)$ 
         $SIMPLEX[3] \leftarrow C$ 
         $ERR[3] \leftarrow C.err$ 
    end if
     $ERR \leftarrow \text{sort}(ERR)$ 
     $SIMPLEX \leftarrow \text{sort}(SIMPLEX) \text{ with } ERR$ 
end while

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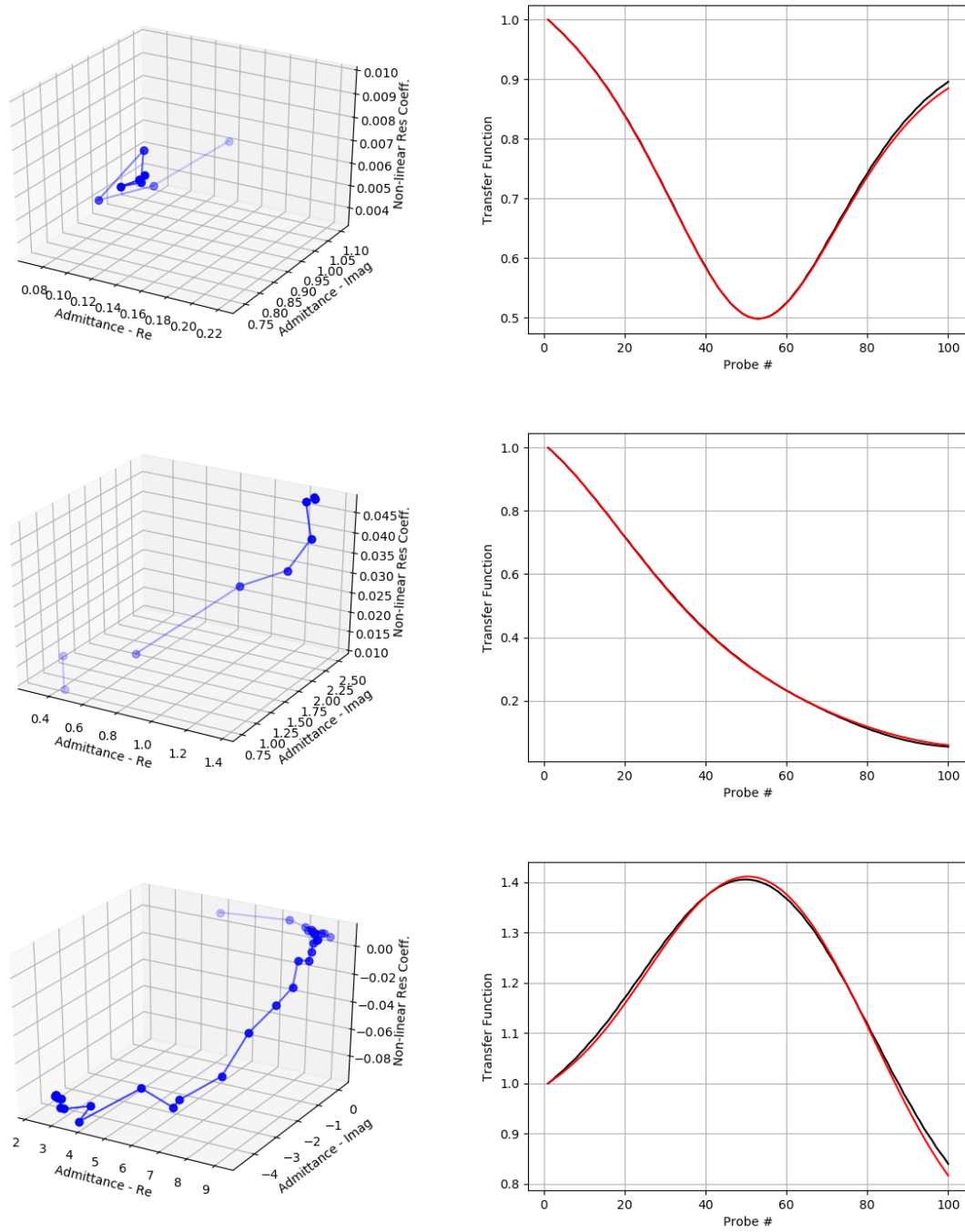


Fig. 7 Optimization parameters convergence history (left) and comparison between the PowerFLOW reference transfer function in black and the one from the reduced order FEM model in red at convergence (right). From top to bottom, wave-packet frequency $\bar{f} = 1000Hz, 1400Hz, 1800Hz$. Cases without grazing flow.

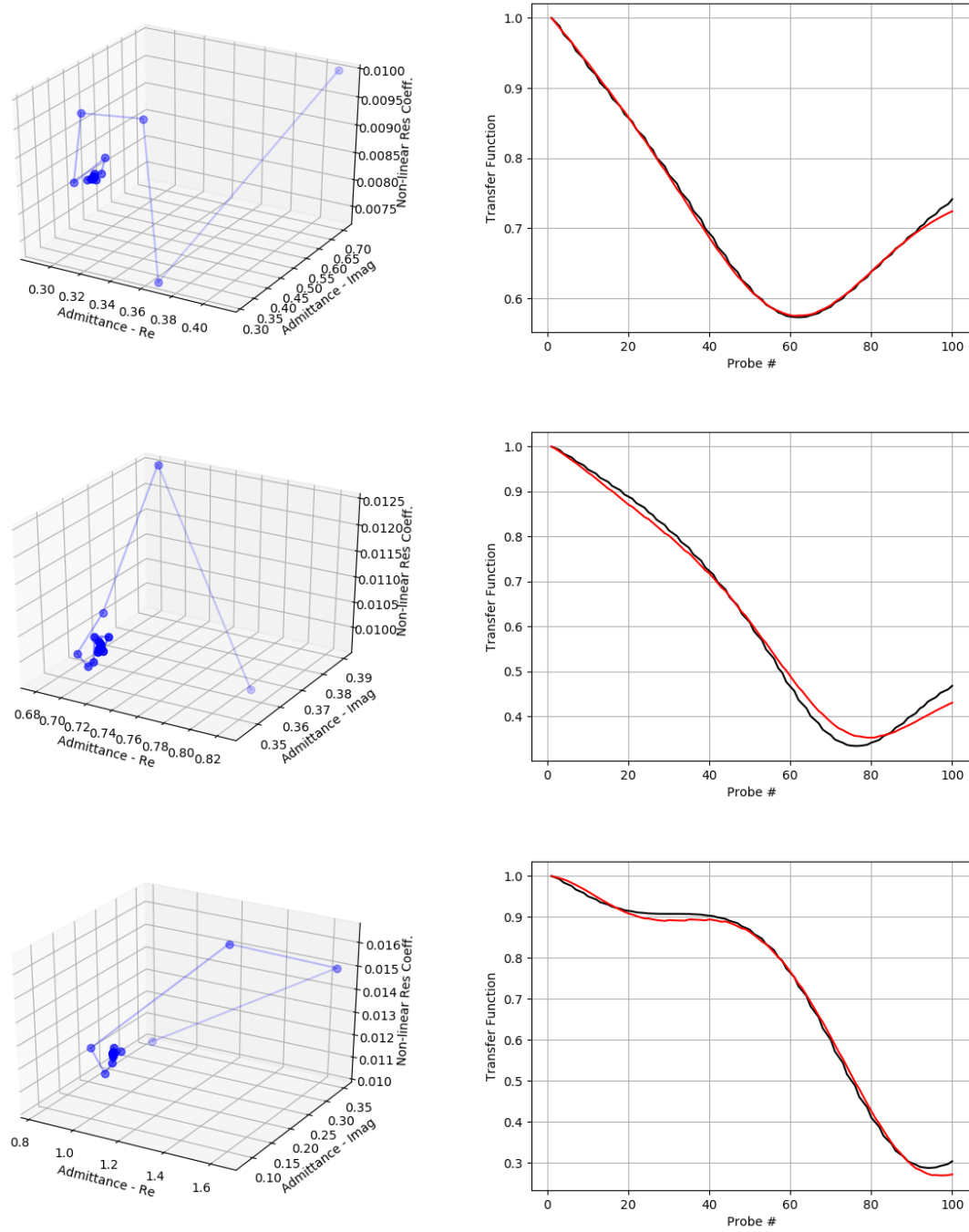


Fig. 8 Optimization parameters convergence history (left) and comparison between the PowerFLOW reference transfer function in black and the one from the reduced order FEM model in red at convergence (right). From top to bottom, wave-packet frequency $\bar{f} = 1000Hz, 1400Hz, 1800Hz$. Cases with grazing flow at $M=0.31$.